Editor

exp135g6g.m | exp5g6g.m | exp155g6g.m | smart.m | untitled.m | ML.m | ML2.m | mlll.m | mageshco1exp1.m | MaheshCo1ex2.m | Maheshco1exp3.m

/MATLAB Drive/Maheshco1exp3.m

```
clear;
clc;
close all;

As = 40;  % Stopband attenuation in dB (not used in this first-order examp
Fs = 5000;
T = 1/Fs;
fc = 1000;
Omega_c = 2*pi*fc;

% Analog Filter Prototype: H(s) = Omega_c / (s + Omega_c)
b_a = [Omega_c];
a_a = [1, Omega_c];

% 1. Impulse Invariance Method
[b_imp, a_imp] = impinvar(b_a, a_a, Fs);

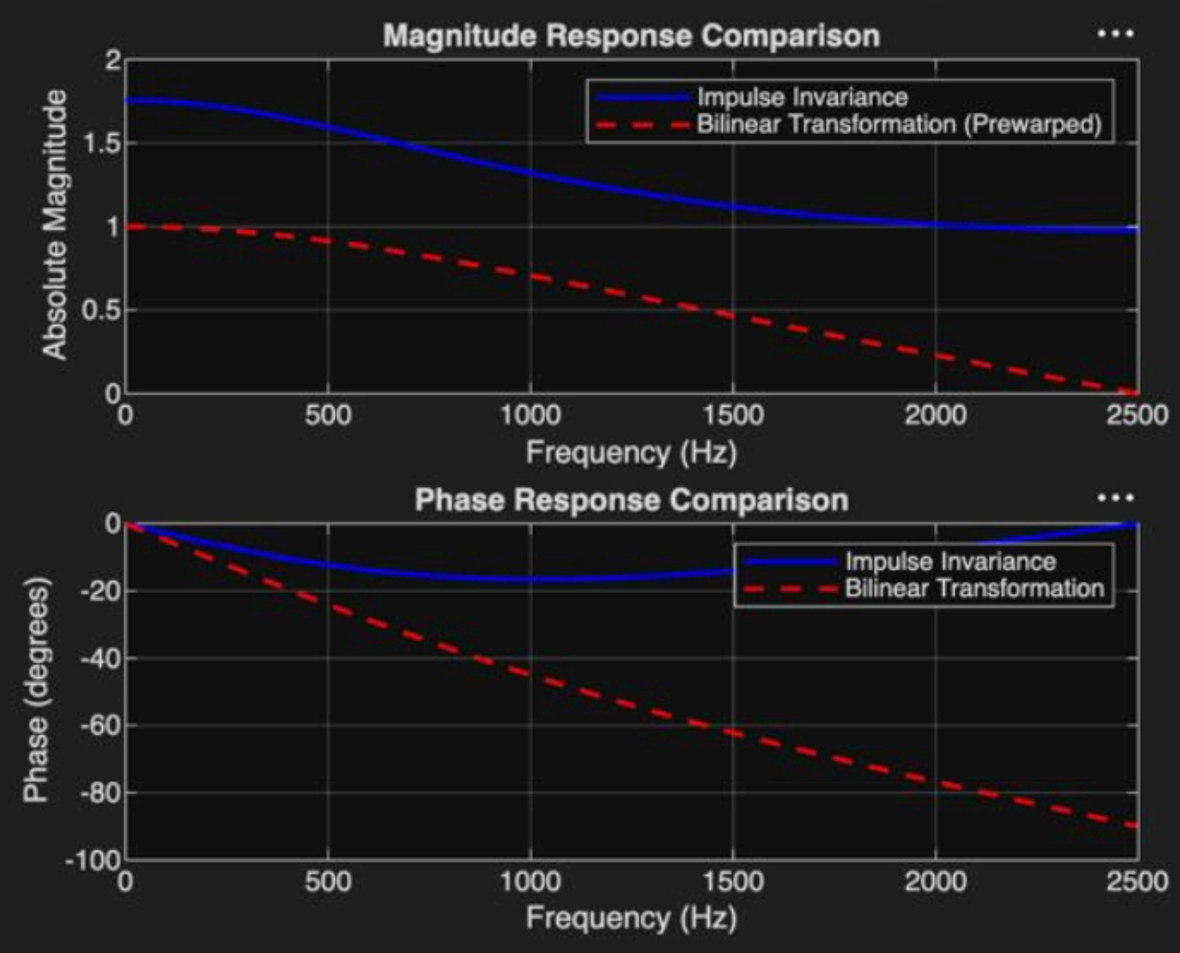
% 2. Bilinear Transformation Method (with Prewarping)
Omega_p = (2/T)*tan(Omega_c*T/2);
b_a_pre = [Omega_p];
a_a_pre = [1, Omega_p];

[b_blt, a_blt] = bilinear(b_a_pre, a_a_pre, Fs);

% Display numerical outputs
fprintf('=== EXPERIMENT 3: IMPULSE INVARIANCE vs BILINEAR TRANSFORMATION =
fprintf('Analog Cutoff Frequency (Omega_c): %.2f rad/s\n', Omega_c);
fprintf('Prewarped Analog Frequency (Omega_p): %.2f rad/s\n\n', Omega_p);
fprintf('--- IMPULSE INVARIANCE COEFFICIENTS ---\n');
fprintf('Digital Numerator (b_imp):\n');
```

Figure 1: Impulse Invariance vs Bilinear Transformation

FIGURE | FORMAT | TOOLS | LAYOUT



Command Window

New to MATLAB? See resources for Getting Started.

```
--- BILINEAR TRANSFORMATION COEFFICIENTS ---
Digital Numerator (b_blt):
    0.4208    0.4208

Digital Denominator (a_blt):
    1.0000   -0.1584

>>
```